

From Softmax Attention to Gated DeltaNet

A derivation from first principles: motivation, mechanism, mathematics, approximations, and limits

THE IDEA IN ONE SENTENCE

Softmax attention already performs an exact prediction-error correction for one fixed query. A local approximation of the newest softmax weight, followed by a heuristic query-independent anchor, turns that correction into a shared matrix update with Gated DeltaNet's retain-erase-write structure.

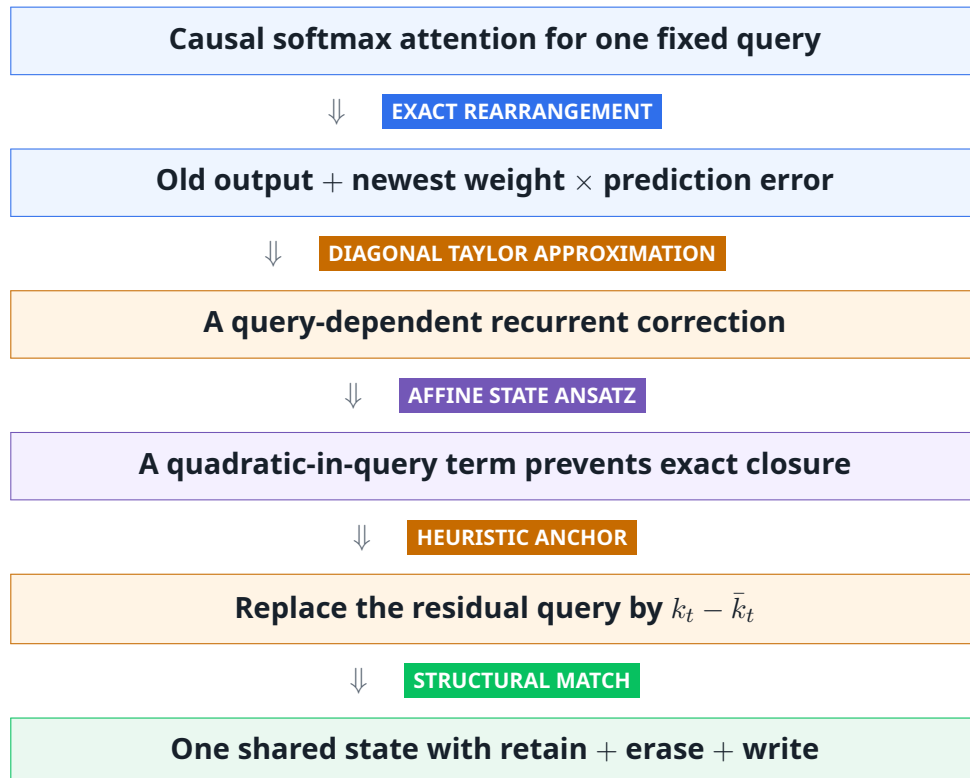
Primary source	Jianlin Su, Scientific Spaces archive 11823 [1]
Reference model	Gated Delta Networks, arXiv:2412.06464 [3]
Format	English, equation-first, editorial explainer
Claim boundary	Analytic reconstruction; no experiment or benchmark claim

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1 The whole argument on one page



READ THIS BEFORE THE EQUATIONS

The conclusion is not “softmax attention equals Gated DeltaNet.” The path contains a local Taylor expansion, an affine modeling assumption, and a norm-dependent heuristic choice. What emerges is a GDN-shaped approximation whose algebra reveals a meaningful relationship between attention and online error correction.

The derivation answers a practical question. Full causal attention can retrieve from every prior token, but it forms a triangular score grid with $1+2+\dots+T$ query–key comparisons. A recurrent model updates one bounded state per token. Can the behavior of softmax attention be compressed into such a state without discarding the mechanism that corrects old associations? The source article’s answer is: approximately, and the correction naturally looks like Gated DeltaNet.

2 Foundations: the objects and operations

2.1 Dimensions and roles

We use one attention head and a causal prefix ending at time t :

$$\mathbf{q}, \mathbf{k}_i \in \mathbb{R}^{d_k}, \quad \mathbf{v}_i, \mathbf{o}_t, \mathbf{b}_t \in \mathbb{R}^{d_v}, \quad \mathbf{A}_t \in \mathbb{R}^{d_v \times d_k}.$$

Object	Shape	Meaning
q	d_k -vector	A request: which stored pattern is relevant now?
k_i	d_k -vector	An address or feature direction for token i .
v_i	d_v -vector	The content returned if key i is selected.
A_t	$d_v \times d_k$ matrix	A compressed map from query directions to value directions.
$\bar{k}_t, \bar{v}_t$	vectors	Arithmetic prefix means through time t .

2.2 Dot product: alignment becomes a scalar

The score $q^\top k$ is a dot product:

$$q^\top k = \sum_{r=1}^{d_k} q_r k_r = \|q\| \|k\| \cos \theta.$$

It is large and positive when the vectors point in similar directions, zero when they are orthogonal, and negative when they oppose each other. Attention uses this scalar as a compatibility score. The usual $1/\sqrt{d_k}$ scaling can be absorbed into q or k for this derivation.

2.3 Softmax: scores become normalized mixture weights

For logits $s_i = q^\top k_i$,

$$\alpha_i = \frac{e^{s_i}}{\sum_j e^{s_j}}, \quad \alpha_i > 0, \quad \sum_i \alpha_i = 1.$$

Exponentiation rewards larger scores; normalization makes the result a convex mixture. If the values are vectors, $\sum_i \alpha_i v_i$ lies in their weighted span.

A SMALL NUMERICAL EXAMPLE

With logits $(-0.2, 0, 0.2)$, exact softmax gives approximately $(0.2693, 0.3289, 0.4018)$. The first-order approximation developed later gives $(0.2667, 0.3333, 0.4000)$. They are close because the logits are tightly clustered; the approximation need not remain close when they spread out.

2.4 Outer product: a key-value association becomes a matrix

The outer product $v k^\top$ has shape $d_v \times d_k$. Reading it with a query gives

$$(v k^\top) q = v (k^\top q).$$

The query first measures its alignment with the stored key, then scales the stored value by that amount. This identity is the algebraic foundation of matrix-state linear attention: a sum of outer products stores many approximate associations in one matrix.

3 Starting point: causal softmax attention

For one fixed query q , attention over the first t key–value pairs is

$$\mathbf{o}_t(q) = \sum_{i=1}^t \alpha_{t,i}(q) \mathbf{v}_i, \quad \alpha_{t,i}(q) = \frac{e^{q^\top \mathbf{k}_i}}{Z_t(q)}, \quad Z_t(q) = \sum_{i=1}^t e^{q^\top \mathbf{k}_i}. \quad (1)$$

SOURCE EQ. 1

DEFINITION Equation (1) says that the query scores every visible key, softmax normalizes those scores, and the resulting weights mix the corresponding values. “Causal” means only positions $1, \dots, t$ are visible at position t .

3.1 The direct first-order route gives vanilla linear attention

Let $\bar{\mathbf{k}}_t = t^{-1} \sum_i \mathbf{k}_i$ and $\bar{\mathbf{v}}_t = t^{-1} \sum_i \mathbf{v}_i$. A first-order expansion of every softmax coefficient gives

$$\begin{aligned} \mathbf{o}_t(q) &\approx \frac{1}{t} \sum_{i=1}^t \left[1 + q^\top (\mathbf{k}_i - \bar{\mathbf{k}}_t) \right] \mathbf{v}_i \\ &= \bar{\mathbf{v}}_t + \left(\frac{1}{t} \sum_{i=1}^t \mathbf{v}_i \mathbf{k}_i^\top - \bar{\mathbf{v}}_t \bar{\mathbf{k}}_t^\top \right) q. \end{aligned} \quad (2)$$

SOURCE EQ. 2

Once the softmax approximation is made, the second line is exact algebra. The matrix is a cross-covariance-like summary of keys and values. This is already a linear-attention state, but it is essentially a Vanilla Linear Attention (VaLA) variant. The source article asks whether a more targeted derivation can retain the Delta Rule’s ability to correct an existing association.

4 The exact Delta-shaped recurrence

Define the unnormalized numerator $N_t = \sum_{i=1}^t e^{q^\top \mathbf{k}_i} \mathbf{v}_i$. Splitting off the newest term gives

$$\mathbf{o}_t = \frac{N_t}{Z_t} = \frac{N_{t-1} + e^{q^\top \mathbf{k}_t} \mathbf{v}_t}{Z_t} = \frac{Z_{t-1} \mathbf{o}_{t-1} + e^{q^\top \mathbf{k}_t} \mathbf{v}_t}{Z_t}. \quad (3)$$

SOURCE EQ. 3

Because $Z_t = Z_{t-1} + e^{q^\top \mathbf{k}_t}$ and $\alpha_{t,t} = e^{q^\top \mathbf{k}_t} / Z_t$,

$$\boxed{\mathbf{o}_t = \mathbf{o}_{t-1} + \alpha_{t,t} (\mathbf{v}_t - \mathbf{o}_{t-1})}. \quad (4)$$

SOURCE EQ. 4

EXACT FOR ONE FIXED QUERY

Equation (4) has the canonical online-learning form

$$\text{new prediction} = \text{old prediction} + \text{step size} \times \text{prediction error}.$$

The error is $v_t - o_{t-1}$ and the adaptive step size is the newest attention weight $\alpha_{t,t}$. No approximation has appeared yet.

4.1 Why this exact RNN is still quadratic

The suppressed dependence matters: the state is really $o_{t-1}(q)$. In a Transformer, each position has a different query $q_1, \dots, q_T$. Query q_t needs a recurrence over its entire prefix, so the amount of work is

$$1 + 2 + \dots + T = \frac{T(T+1)}{2} = \mathcal{O}(T^2).$$

An actually linear recurrent model needs one state that is updated only from keys and values, independent of whichever query will read it later. The rest of the derivation constructs an approximation to that shared state.

5 The local softmax approximation**5.1 First-order expansion from the ground up**

Let $s_i = q^\top k_i$, let $\bar{s} = t^{-1} \sum_i s_i$, and define centered logits $\delta_i = s_i - \bar{s}$. Then $\sum_i \delta_i = 0$. For small δ_i ,

$$e^{s_i} = e^{\bar{s}} e^{\delta_i} = e^{\bar{s}} \left(1 + \delta_i + \frac{\delta_i^2}{2} + \dots \right).$$

At first order the denominator is

$$\sum_j e^{s_j} \approx e^{\bar{s}} \sum_j (1 + \delta_j) = t e^{\bar{s}},$$

because the centered deviations cancel. Therefore

$$\alpha_{t,i}(q) \approx \frac{1}{t} (1 + \delta_i) = \frac{1}{t} \left[1 + q^\top (k_i - \bar{k}_t) \right]. \quad (\text{T})$$

This approximation still sums to one, but it no longer guarantees positivity. Its omitted error begins at second order in the spread of the centered logits. It is a local expansion around uniform attention, not a global replacement for softmax [2].

5.2 Approximate only the newest coefficient

Apply Equation (T) only to $\alpha_{t,t}$ in the exact recurrence. Writing $\delta k_t = k_t - \bar{k}_t$ gives

$$\begin{aligned}
\mathbf{o}_t &\approx \mathbf{o}_{t-1} + \frac{1}{t} \left(1 + \mathbf{q}^\top \delta \mathbf{k}_t \right) (\mathbf{v}_t - \mathbf{o}_{t-1}) \\
&= \left(1 - \frac{1}{t} \right) \mathbf{o}_{t-1} + \frac{1}{t} \mathbf{v}_t + \frac{1}{t} (\mathbf{v}_t - \mathbf{o}_{t-1}) \delta \mathbf{k}_t^\top \mathbf{q}.
\end{aligned} \tag{5}$$

SOURCE EQ. 5

LOCAL APPROXIMATION This is a lighter approximation burden than expanding all $\alpha_{t,i}$ at once, although it is not automatically accurate for every sequence.

5.3 Why the diagonal route can retain more structure

For a fixed query, every old weight has the exact factorization

$$\alpha_{t,i} = \alpha_{i,i} \prod_{j=i+1}^t (1 - \alpha_{j,j}). \tag{6}$$

SOURCE EQ. 6

To see this, insert the telescoping normalizers:

$$\frac{e^{s_i}}{Z_t} = \frac{e^{s_i}}{Z_i} \frac{Z_i}{Z_{i+1}} \cdots \frac{Z_{t-1}}{Z_t}, \quad \frac{Z_{j-1}}{Z_j} = 1 - \frac{e^{s_j}}{Z_j} = 1 - \alpha_{j,j}.$$

Approximating the local factors and multiplying them keeps cross-products such as $(1 + a)(1 + b) = 1 + a + b + ab$. A single global first-order expansion would keep only $1 + a + b$. This motivates the diagonal strategy, but it is not a theorem that the accumulated approximation error is uniformly smaller.

6 A shared affine memory, and why it fails to close

The smallest useful query-independent model is an affine readout:

$$\hat{\mathbf{o}}_t(\mathbf{q}) = \mathbf{A}_t \mathbf{q} + \mathbf{b}_t, \quad \mathbf{A}_0 = 0, \quad \mathbf{b}_0 = 0. \tag{7}$$

SOURCE EQ. 7

MODELING ANSATZ The matrix and bias may depend on all observed keys and values, but they must not depend on the query used at readout time. This is deliberately an approximation: generic softmax attention is not affine in $\mathbf{q}$.

Substitute Equation (7) into Equation (5):

$$\begin{aligned}
\mathbf{A}_t \mathbf{q} + \mathbf{b}_t &\approx \left(1 - \frac{1}{t} \right) (\mathbf{A}_{t-1} \mathbf{q} + \mathbf{b}_{t-1}) + \frac{1}{t} \mathbf{v}_t \\
&\quad + \frac{1}{t} (\mathbf{v}_t - \mathbf{b}_{t-1} - \mathbf{A}_{t-1} \mathbf{q}) \delta \mathbf{k}_t^\top \mathbf{q}.
\end{aligned} \tag{8}$$

SOURCE EQ. 8

The query-free part closes cleanly:

$$\mathbf{b}_t = \left(1 - \frac{1}{t}\right) \mathbf{b}_{t-1} + \frac{1}{t} \mathbf{v}_t = \bar{\mathbf{v}}_t. \quad (9)$$

SOURCE EQ. 9

But the apparent matrix update is

$$\mathbf{A}_t = \left(1 - \frac{1}{t}\right) \mathbf{A}_{t-1} + \frac{1}{t} (\mathbf{v}_t - \bar{\mathbf{v}}_{t-1} - \mathbf{A}_{t-1} \mathbf{q}) \delta \mathbf{k}_t^\top. \quad (10)$$

SOURCE EQ. 10

The visible $\mathbf{q}$ violates query independence. Equivalently, Equation (8) contains

$$-(\mathbf{A}_{t-1} \mathbf{q})(\delta \mathbf{k}_t^\top \mathbf{q}),$$

which is quadratic in $\mathbf{q}$, while the left side is only affine. This is the exact obstruction: first-order softmax does not imply first-order closure of the recurrent state.

7 The shortcut that only returns to VaLA

One could simply drop $\mathbf{A}_{t-1} \mathbf{q}$ inside the residual, obtaining

$$\mathbf{A}_t = \left(1 - \frac{1}{t}\right) \mathbf{A}_{t-1} + \frac{1}{t} (\mathbf{v}_t - \bar{\mathbf{v}}_{t-1})(\mathbf{k}_t - \bar{\mathbf{k}}_t)^\top. \quad (11)$$

SOURCE EQ. 11

Because this is an incremental average, it unrolls exactly to

$$\mathbf{A}_t = \frac{1}{t} \sum_{i=1}^t (\mathbf{v}_i - \bar{\mathbf{v}}_{i-1})(\mathbf{k}_i - \bar{\mathbf{k}}_i)^\top. \quad (12)$$

SOURCE EQ. 12

This looks new, but two exact mean identities reveal that it is merely the VaLA matrix in Equation (2). First,

$$t \bar{\mathbf{v}}_t \bar{\mathbf{k}}_t^\top - (t-1) \bar{\mathbf{v}}_{t-1} \bar{\mathbf{k}}_{t-1}^\top = -\bar{\mathbf{v}}_{t-1} \bar{\mathbf{k}}_t^\top + \mathbf{v}_t \bar{\mathbf{k}}_t^\top + \bar{\mathbf{v}}_{t-1} \mathbf{k}_t^\top. \quad (13)$$

SOURCE EQ. 13

Summing this telescoping difference gives

$$\sum_{i=1}^t \mathbf{v}_i \mathbf{k}_i^\top - t \bar{\mathbf{v}}_t \bar{\mathbf{k}}_t^\top = \sum_{i=1}^t (\mathbf{v}_i - \bar{\mathbf{v}}_{i-1})(\mathbf{k}_i - \bar{\mathbf{k}}_i)^\top. \quad (14)$$

SOURCE EQ. 14

WHY THIS NEGATIVE RESULT MATTERS

Throwing away the troublesome prediction term also throws away the mechanism that erases an obsolete key–value association. The derivation has gone in a circle and recovered vanilla linear attention. To obtain a Delta-style state, the old prediction must be approximated more carefully rather than deleted.

8 The error principle and the representative query

Restore the query dependence explicitly in the exact fixed-query recurrence:

$$\mathbf{o}_t(\mathbf{q}) = \mathbf{o}_{t-1}(\mathbf{q}) + \alpha_{t,t}(\mathbf{q})[\mathbf{v}_t - \mathbf{o}_{t-1}(\mathbf{q})]. \quad (15)$$

SOURCE EQ. 15

Suppose only the occurrence inside the residual is replaced by a representative query $\mathbf{c}$. The introduced error is exactly

$$\mathbf{E}_t(\mathbf{q}, \mathbf{c}) = \alpha_{t,t}(\mathbf{q})[\mathbf{o}_{t-1}(\mathbf{q}) - \mathbf{o}_{t-1}(\mathbf{c})]. \quad (16)$$

SOURCE EQ. 16

Choosing $\mathbf{c} = \mathbf{q}_t$ eliminates today's error at one query, but the shared state must also serve unknown future queries. The article proposes a minimax-inspired anchor: make the replacement exact at the query direction where the newest token would receive its largest attention weight.

Under the linearized weight,

$$\alpha_{t,t}(\mathbf{q}) \approx \frac{1}{t} [1 + \mathbf{q}^\top \delta \mathbf{k}_t].$$

Without a norm constraint this has no finite maximum. If $\|\mathbf{q}\| \leq r$, Cauchy–Schwarz gives

$$\mathbf{q}^\top \delta \mathbf{k}_t \leq \|\mathbf{q}\| \|\delta \mathbf{k}_t\| \leq r \|\delta \mathbf{k}_t\|,$$

with equality in the direction $r \delta \mathbf{k}_t / \|\delta \mathbf{k}_t\|$. Assuming comparable query and key norms, the source makes the simple magnitude choice

$$\mathbf{q}^* = \delta \mathbf{k}_t = \mathbf{k}_t - \bar{\mathbf{k}}_t. \quad (17)$$

SOURCE EQ. 17

WHAT IS RIGOROUS, AND WHAT IS HEURISTIC?

The maximizing *direction* follows from Cauchy–Schwarz once a query-norm bound is imposed. The magnitude $\mathbf{q}^* = \delta \mathbf{k}_t$ additionally assumes comparable query/key scales. This does not solve a global minimax problem for Equation (16); it anchors the approximation at one strategically selected direction.

9 The Gated DeltaNet-shaped update

Define the centered key and value innovation

$$\mathbf{x}_t = \mathbf{k}_t - \bar{\mathbf{k}}_t, \quad \mathbf{y}_t = \mathbf{v}_t - \bar{\mathbf{v}}_{t-1}, \quad g_t = \frac{1}{t}, \quad \lambda_t = 1 - \frac{1}{t}.$$

Substituting $\mathbf{q}^* = \mathbf{x}_t$ into Equation (10) yields

$$\mathbf{A}_t = \lambda_t \mathbf{A}_{t-1} + g_t (\mathbf{y}_t - \mathbf{A}_{t-1} \mathbf{x}_t) \mathbf{x}_t^\top \quad (18a)$$

$$= \mathbf{A}_{t-1} (\lambda_t \mathbf{I} - g_t \mathbf{x}_t \mathbf{x}_t^\top) + g_t \mathbf{y}_t \mathbf{x}_t^\top. \quad (18b)$$

SOURCE EQ. 18

RETAIN $\lambda_t \mathbf{A}_{t-1}$ Keep a decayed copy of global memory.	ERASE $-g_t (\mathbf{A}_{t-1} \mathbf{x}_t) \mathbf{x}_t^\top$ Remove the old prediction along $\mathbf{x}_t$.	WRITE $g_t \mathbf{y}_t \mathbf{x}_t^\top$ Store the new centered target at $\mathbf{x}_t$.
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9.1 The fundamental Delta Rule from gradient descent

Suppose a matrix S should map key $\mathbf{k}$ to target value v . Use squared prediction loss

$$L(S) = \frac{1}{2} \|\mathbf{v} - S\mathbf{k}\|^2.$$

The matrix gradient is

$$\nabla_S L = (S\mathbf{k} - \mathbf{v})\mathbf{k}^\top.$$

One gradient-descent step with learning rate β is

$$\begin{aligned} S^+ &= S - \beta (S\mathbf{k} - \mathbf{v})\mathbf{k}^\top \\ &= S + \beta (\mathbf{v} - S\mathbf{k})\mathbf{k}^\top \\ &= S(\mathbf{I} - \beta \mathbf{k}\mathbf{k}^\top) + \beta \mathbf{v}\mathbf{k}^\top. \end{aligned} \quad (D)$$

The term $S\mathbf{k}$ is the old prediction; the residual $\mathbf{v} - S\mathbf{k}$ is what remains to be learned. Multiplication by $\mathbf{k}^\top$ localizes the matrix correction to that key direction. Equation (18a) is this targeted correction plus a global retention factor λ_t .

9.2 A concrete two-dimensional update

Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0.5 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, \quad \beta = 0.25.$$

The old prediction is $\mathbf{A}\mathbf{x} = (2, 0)^\top$ and the residual is $(1, -1)^\top$. A plain Delta step gives

$$\mathbf{A}^+ = \mathbf{A} + 0.25 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 2.25 & 0.5 \\ -0.25 & 1 \end{bmatrix}.$$

Only the first column changes because $\mathbf{x}$ points along the first input axis. The updated prediction $(2.25, -0.25)^\top$ has moved one quarter of the way from the old prediction toward the target. This is the geometric meaning of the rank-one erase/write operation.

10 Comparison with canonical Gated DeltaNet

A common Gated DeltaNet convention is [3]

$$\mathbf{S}_t = \mathbf{S}_{t-1}[\alpha_t(\mathbf{I} - \beta_t \mathbf{k}_t \mathbf{k}_t^\top)] + \beta_t \mathbf{v}_t \mathbf{k}_t^\top. \quad (19)$$

SOURCE EQ. 19

Both updates have the same three mechanisms: global retention, targeted erasure of the current key's old prediction, and a rank-one write. For $t > 1$, Equation (18b) can even be placed into the convention of Equation (19) by the algebraic reparameterization

$$\alpha_t = \frac{t-1}{t}, \quad \beta_t = \frac{1}{t-1}, \quad \mathbf{k}'_t = \mathbf{x}_t, \quad \mathbf{v}'_t = \frac{t-1}{t} \mathbf{y}_t.$$

Indeed, $\alpha_t \beta_t = 1/t$ and $\beta_t \mathbf{v}'_t = \mathbf{y}_t/t$. At $t = 1$, the centered key is zero and the state can be initialized directly rather than using the singular reparameterization.

	Taylor-derived state	Trained Gated DeltaNet
Retention	Deterministic $1 - 1/t$	Learned, data-dependent gate
Update rate	Deterministic $1/t$	Learned, data-dependent gate
Key	Centered $\mathbf{k}_t - \bar{\mathbf{k}}_t$	Learned key projection
Value target	$\mathbf{v}_t - \bar{\mathbf{v}}_{t-1}$	Learned value projection
Status	Approximation to softmax	Independently trained architecture
Evidence here	Analytic structure	No benchmark transferred by this derivation

THE PRECISE CLAIM

There is an exact algebraic mapping between the displayed update forms after centering and rescaling their variables. There is not an equality between ordinary softmax attention and a trained Gated DeltaNet model. The latter learns its representations and gates from data, whereas the derivation fixes them through Taylor coefficients and prefix means.

11 Readout, algorithm, and complexity

The approximate output for the actual query at time t is

$$\hat{o}_t = A_t q_t + \bar{v}_t. \quad (20)$$

SOURCE EQ. 20

The hat is intentional: it keeps visible that this is the output of the approximating recurrent model, not an exact identity with the original softmax output.

11.1 One causal scan

Step	Operation	Purpose
1	$g_t = 1/t, \lambda_t = 1 - g_t$	Set deterministic write and retention rates.
2	$\bar{k}_t = \lambda_t \bar{k}_{t-1} + g_t k_t$	Update the key mean.
3	$x_t = k_t - \bar{k}_t$	Isolate what is new about the key.
4	$y_t = v_t - \bar{v}_{t-1}$	Isolate the value innovation against the old baseline.
5	$A_t = \lambda_t A_{t-1} + g_t (y_t - A_{t-1} x_t) x_t^\top$	Retain globally, erase the old prediction, and write the residual.
6	$\bar{v}_t = \lambda_t \bar{v}_{t-1} + g_t v_t$	Update the value baseline.
7	$\hat{o}_t = A_t q_t + \bar{v}_t$	Read the shared state with today's query.

At $t = 1$, $\bar{k}_1 = k_1$, so $x_1 = 0$, $A_1 = 0$, and the output is $\bar{v}_1 = v_1$, exactly as it should be for a one-token prefix.

11.2 Sequence scaling

	Causal softmax attention	Derived recurrent approximation
Sequence state	All prior keys and values: $\mathcal{O}(T(d_k + d_v))$	Matrix and two means: $\mathcal{O}(d_k d_v + d_k + d_v)$
Per-token work	Grows with prefix length	$\mathcal{O}(d_k d_v)$ rank-one update/read
Full sequence	$\mathcal{O}(T^2 d_k)$ score interactions, plus value mixing	$\mathcal{O}(T d_k d_v)$
Scaling statement	Quadratic in T	Linear in T for fixed head dimensions

Linear asymptotic sequence cost does not guarantee faster wall-clock execution. Kernel fusion, parallel training scans, memory traffic, head size, and hardware utilization all matter. The source article provides no timing measurements.

12 Assumption and claim ledger

Class	Where	Boundary
EXACT	Equations 3–4	Exact recurrence for one fixed query and its prefix.
EXACT	Equation 6	Exact factorization of old attention weights into diagonal survival factors.
TAYLOR	Equations 2 and 5	Local around nearly uniform logits; may yield negative approximate weights.
ANSATZ	Equation 7	A finite affine state cannot represent generic softmax dependence on every query exactly.
HEURISTIC	Equation 17	Direction follows under a norm constraint; magnitude assumes comparable query/key scales.
STRUCTURE	Equations 18–19	Retain–erase–write forms match after centering/rescaling; learned models are not thereby equal.
NOT CLAIMED	Performance	The article contains no experiment, benchmark, accuracy guarantee, or universal replacement claim.

Additional boundaries matter in practice:

- Repeated local approximation errors may accumulate through the scan.
- The minimax-inspired anchor does not control the replacement error for every possible future query.
- The derivation assumes causal dot-product attention. Extra biases and masks need separate analysis, as do positional terms and nonlinear feature maps.
- The deterministic $1/t$ write rate becomes small on long prefixes; learned GDN gates need not follow this schedule.
- Centered keys can have varying norms, so practical implementations may need normalization or learned scaling not justified by this derivation.

13 Research ideas suggested by the mechanism

The following are testable follow-ups, not claims made by the source article.

13.1 Use logit dispersion as an approximation-confidence signal

The Taylor error grows with centered-logit spread. A model could estimate $\text{Var}_i(\mathbf{q}_i^\top \mathbf{k}_i)$ or a recurrent proxy and use it to interpolate between the cheap Delta state and a local attention fallback. This directly tests whether the mathematical validity region predicts model error.

13.2 Initialize learned gates from the harmonic schedule

The derivation suggests $g_t = 1/t$ and $\lambda_t = 1 - 1/t$ as an analytic starting point. A learned GDN could be initialized near this schedule, then trained to depart from it when the data calls for faster writing or forgetting. The useful comparison is initialization and stability, not an assumption that the fixed schedule is optimal.

13.3 Ablate centering separately from the Delta Rule

The derived key is $\mathbf{k}_t - \bar{\mathbf{k}}_t$ and the value target is $\mathbf{v}_t - \bar{\mathbf{v}}_{t-1}$. A controlled experiment should compare uncentered, key-centered, value-centered, and doubly centered updates while holding parameter count and gate parameterization fixed.

13.4 Measure diagonal approximation error over time

Equation (6) provides an interpretable diagnostic: compare exact diagonal weights, their Taylor approximations, and the product-induced old weights. This can reveal whether the local-factor strategy really outperforms a single global expansion in trained representations, and where errors compound.

13.5 Benchmark quality and systems behavior separately

Any evaluation should report approximation error, language-model quality, long-context retrieval, recurrent-state memory, training throughput, and single-token decoding latency. Linear asymptotics and hardware speed are different hypotheses.

FINAL TAKEAWAY

Softmax contains an exact Delta-shaped update when the query is fixed. The article linearizes only the newest attention gate, exposes the query term that prevents a shared state, and anchors that term to the centered newest key. The resulting matrix recurrence retains old memory, erases its prediction in one key direction, and writes the residual there. That is why Gated DeltaNet's mechanism appears—approximately and structurally, not by exact model equivalence.

A Dimension check for the final update

For $\mathbf{A}_{t-1} \in \mathbb{R}^{d_v \times d_k}$, $\mathbf{x}_t \in \mathbb{R}^{d_k}$, and $\mathbf{y}_t \in \mathbb{R}^{d_v}$:

$$\underbrace{\mathbf{A}_{t-1}\mathbf{x}_t}_{d_v\text{-vector}}, \quad \underbrace{\mathbf{y}_t - \mathbf{A}_{t-1}\mathbf{x}_t}_{d_v\text{-vector}}, \quad \underbrace{(\mathbf{y}_t - \mathbf{A}_{t-1}\mathbf{x}_t)\mathbf{x}_t^\top}_{d_v \times d_k \text{ matrix}}.$$

Every term in Equation (18a) therefore has the same $d_v \times d_k$ shape. Reading with $\mathbf{q}_t \in \mathbb{R}^{d_k}$ produces a d_v -vector, which can be added to $\bar{\mathbf{v}}_t$.

B Source-equation map

Source	Role here	Status
1–2	Attention definition and global first-order VaLA baseline	Definition, then local approximation
3–4	Prefix split and prediction-error recurrence	Exact for a fixed query
5–6	Diagonal Taylor step and survival factorization	Approximation, then exact identity
7–10	Affine state and quadratic-query obstruction	Modeling ansatz and inherited approximation
11–14	Dropped-query shortcut and return to VaLA	Heuristic followed by exact telescoping algebra
15–17	Replacement error and representative direction	Exact error expression followed by heuristic choice
18–20	GDN-shaped update, canonical comparison, and readout	Derived approximation and structural comparison

References

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- [3] Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving mamba2 with delta rule. *arXiv preprint arXiv:2412.06464*, 2024.